

# The Business, Tax, and Financial Environments

# Objectives

- To understand better the role of financial managers, you must be familiar with the environments in which they operate.
- The form of business organization that a firm chooses is one aspect of the business setting in which it must function.
- We will explore the advantages and disadvantages of the various alternative forms of business organization

# Types of Environments

- The Business Environment
- The Tax Environment
- The Financial Environment

# The Business Environment

- there are four basic forms of business organization:
  - sole proprietorships (one owner),
  - partnerships (general and limited),
  - corporations, and
  - limited liability companies (LLCs).

# Sole Proprietorships

- A business form for which there is one owner. This single owner has unlimited liability for all debts of the firm.
- The **sole proprietorship** is the oldest form of business organization.
- As the title suggests, a single person owns the business, holds title to all its assets, and is personally responsible for all of its debts.
- A proprietorship pays no separate income taxes. The owner merely adds any profits or subtracts any losses from the business when determining personal taxable income.

# Sole Proprietorship

| Advantages                           | Disadvantages                      |
|--------------------------------------|------------------------------------|
| Simplicity                           | Unlimited liability                |
| Low setup cost                       | Hard to raise additional capital   |
| Quick setup                          | Transfer of ownership difficulties |
| Single tax filing on individual form |                                    |

# Partnership

- A business form in which two or more individuals act as owners.
- In a *general partnership* all partners have unlimited liability for the debts of the firm
- they are jointly liable for the obligations of the partnership. Because each partner can bind the partnership with obligations general partners should be selected with care.
- in a *limited partnership* one or more partners may have limited liability.
- There must, however, be at least one **general partner** in the partnership, whose liability is unlimited. Limited partners do not participate in the operation of the business; this is left to the general partner(s).

# Partnership

| Advantages                                      | Disadvantages                                                              |
|-------------------------------------------------|----------------------------------------------------------------------------|
| Can be simple                                   | Unlimited liability for the general partner                                |
| Low setup cost, higher than sole proprietorship | Difficult to raise additional capital, but easier than sole proprietorship |
| Relatively quick setup                          | Transfer of ownership difficulties                                         |
| Limited liability for limited partners          |                                                                            |



# Corporation

- A business form legally separate from its owners.
- A **corporation** is an “artificial entity” created by law. It can own assets and incur liabilities.
- An owner’s liability is limited to his or her investment.
- Ownership itself is evidenced by shares of stock, with each stockholder owning that proportion of the enterprise represented by his or her shares in relation to the total number of shares outstanding.
- Business income is accounted for on the income tax form of the corporation.

# Corporation

| Advantages                                  | Disadvantages                         |
|---------------------------------------------|---------------------------------------|
| Limited liability                           | Double taxation                       |
| Easy transfer of ownership                  | More difficult to establish           |
| Unlimited life                              | More expensive to set up and maintain |
| Easier to raise large quantities of capital |                                       |
|                                             |                                       |

# Limited Liability Companies (LLCs)

- A business form that provides its owners (called “members”) with corporate-style limited personal liability and the federal-tax treatment of a partnership.
- is a hybrid form of business organization that combines the best aspects of both a corporation and a partnership.
- Business income is accounted for on each “member’s” individual income tax form.

# Limited Liability Companies (LLCs)

- Limited liability companies generally possess no more than two of the following four (desirable) standard corporate characteristics:
  - (1) limited liability,
  - (2) centralized management,
  - (3) unlimited life, and
  - (4) the ability to transfer ownership interest without prior consent of the other owners.

# Limited Liability Companies (LLCs)

| Advantages                                 | Disadvantages                                  |
|--------------------------------------------|------------------------------------------------|
| Limited liability                          | Limited life (generally)                       |
| Eliminates double taxation                 | Transfer of ownership difficulties (generally) |
| No restriction on number or type of owners |                                                |
| Easier to raise additional capital         |                                                |

# The Tax Environment

- Most business decisions are affected either directly or indirectly by taxes.
- Through their taxing power, federal, state, and local governments have a profound influence on the behavior of businesses and their owners.

# Corporate Income Taxes

- A corporation's taxable income is found by deducting all allowable expenses, including depreciation and interest, from revenues.

| Company type                                                                           | Tax rate (%) |
|----------------------------------------------------------------------------------------|--------------|
| Banking company                                                                        | 39           |
| Public company* other than a banking company                                           | 29           |
| Any other company                                                                      | 29           |
| Small company (see the <i>Tax credits and incentives</i> section for more information) | 20           |

# Corporate Income Taxes

| Income (PKR*) |             | Super tax rate (%) |
|---------------|-------------|--------------------|
| Over          | Not over    |                    |
| 150 million   | 200 million | 1                  |
| 200 million   | 250 million | 1.5                |
| 250 million   | 300 million | 2.5                |
| 300 million   | 350 million | 3.5                |
| 350 million   | 400 million | 5.5                |
| 400 million   | 500 million | 7.5                |
| 500 million   |             | 10                 |



# Corporate Income Taxes

- The tax rate – the percentage of taxable income that must be paid in taxes – that is applied to each income bracket is referred to as a *marginal rate*.
- The *average tax rate* for a firm is measured by dividing taxes actually paid by taxable income.

# Income Tax Example

- Lisa (Pvt) Ltd. is calculating the income tax liability, marginal tax rate, and average tax rate for the fiscal year ending December 31.
- Lisa (Pvt) Ltd. taxable income for this fiscal year was Rs. 250M.

$$\text{Income tax liability} = 250M \times (0.29) = 72,500,000$$

$$\text{Super Tax} = 250M \times (0.025) = 6,250,000$$

- Marginal Tax Rate is 29%
- Average tax rate =  $78,750,000 / 250 \text{ M}$   
= **31.5%**

# Other Tax Issues

- Alternative Minimum Tax is a special tax which equals 20% of alternative minimum taxable income (generally not equal to taxable income). Corporations pay the maximum of AMT or regular tax liability.
- Quarterly Tax Payments require corporations to pay 25% of their estimated annual tax liability on the 15th of April, June, September, and December.

# Depreciation

- **Depreciation** is the systematic allocation of the cost of a capital asset over a period of time for financial reporting purposes, tax purposes, or both.
- Depreciation deductions taken on a firm's tax return are treated as expense items. Thus depreciation lowers taxable income.
- Everything else being equal, the greater the depreciation charges, the lower the tax.
- Generally, profitable firms prefer to use an accelerated method for tax reporting purposes.

# Common Types of Depreciation

- Straight-line (SL)
- A method of depreciation that allocates expenses evenly over the depreciable life of the asset.
- Accelerated Types
- **Declining-balance depreciation**, on the other hand, calls for an annual charge that is a “fixed percentage” of the asset’s net book value (acquisition cost minus accumulated depreciation) at the beginning of the year to which the depreciation charge applies.
- **Double Declining Balance (DDB)**
- When using the *double-declining-balance (DDB) method*, we compute a rate by dividing 1 by the number of years of depreciable life for the asset. Then we double that rate.
- Under the declining-balance methods the general formula for determining the depreciation charge in any period is

$$m(1/n)NBV$$

# Modified Accelerated Cost Recovery System (MACRS)

- The [Modified Accelerated Cost Recovery System \(MACRS\)](#) is the current tax depreciation system in the U.S. for assets placed in service after 1986, allowing businesses to recover capitalized costs over specified periods. It utilizes accelerated methods, providing larger deductions in the early years of an asset's life
- For the 3-, 5-, 7-, and 10-year property classes, the double declining balance (also called 200% declining balance) depreciation method is used. This method then switches to straight-line depreciation for the remaining undepreciated book value in the first year that the straight-line method yields an equal or greater deduction than the declining-balance method.
- Assets in the 15- and 20-year classes are depreciated using the 150 percent declining balance method, again switching to straight-line at the optimal time. The straight-line method must be used for all real estate.

# Modified Accelerated Cost Recovery System (MACRS)

- Normally, the *half-year convention* must be applied to all declining-balance methods. This calls for a half year of depreciation in the year an asset is acquired, regardless of the date of purchase. There is also a half year of depreciation in the year an asset is sold or retired from service.
- Assets are depreciated based on one of eight different property classes.
- Assets are assigned to specific recovery periods (e.g., 3, 5, 7, 10, 15, 20, 27.5, or 39 years) based on type.

# Modified Accelerated Cost Recovery System (MACRS) (Assets Class)

- *3-Year 200% Class.* Includes property with a midpoint life of 4 years or less, except automobiles and light trucks. Under the Asset Depreciation Range (ADR) system, assets are grouped within classes and a guideline (midpoint) life is determined by the Treasury Department.
- *5-Year 200% Class.* Includes property with an ADR midpoint life of more than 4 to less than 10 years. Also included are automobiles, light trucks, most technological and semiconductor manufacturing equipment, switching equipment, small power production facilities, research and experimental equipment, high technology medical equipment, computers, and certain office equipment.
- *7-Year 200% Class.* Includes property with ADR midpoints of 10 to less than 16 years and single-purpose agricultural structures. Also includes office furniture and any other property for which no class life is specified by law.
- *10-Year 200% Class.* Includes property with ADR midpoints of 16 to less than 20 years.
- *15-Year 150% Class.* Includes property with ADR midpoints of 20 to less than 25 years, sewage treatment plants, and telephone distribution plants.
- *20-Year 150% Class.* Includes property with ADR midpoints of 25 years or more, other than real property described below.
- *27.5-Year Straight-Line Class.* Includes residential rental property.
- *39-Year Straight-Line Class.* Includes other real estate.



# Modified Accelerated Cost Recovery System (MACRS) (Assets Class)

| Recovery Period | Property Type                                     | Common Examples                                                                  |
|-----------------|---------------------------------------------------|----------------------------------------------------------------------------------|
| 3-year          | Property with class life < 4 years                | Tractors, racehorses (over 2 years old), qualified rent-to-own property 8        |
| 5-year          | Property with class life $\geq 4$ but < 10 years  | Automobiles, buses, taxis, computers, office equipment, construction machinery 8 |
| 7-year          | Property with class life $\geq 10$ but < 16 years | Office furniture, fixtures, appliances, most machinery and equipment 1           |
| 10-year         | Property with class life $\geq 16$ but < 20 years | Vessels, barges, tugs, certain manufacturing equipment 7                         |
| 15-year         | Property with class life $\geq 20$ but < 25 years | Land improvements (fences, paving), certain utility property 9                   |
| 20-year         | Property with class life $\geq 25$ years          | Farm buildings, certain municipal sewers 9                                       |
| 27.5-year       | Residential rental property                       | Buildings where 80%+ of gross rental income comes from dwelling units 1          |
| 39-year         | Nonresidential real property                      | Office buildings, stores, warehouses, commercial structures 1                    |

# Depreciation Example

- Lisa Miller of Basket Wonders (BW) is calculating the depreciation on a machine with a depreciable basis of \$100,000, a 6-year useful life, and a 5-year property class life.
- She calculates the annual depreciation charges using MACRS.

# Depreciation Example

| YEAR | DEPRECIATION<br>CALCULATION | DEPRECIATION<br>CHARGE | NET BOOK VALUE<br>(END OF YEAR) |
|------|-----------------------------|------------------------|---------------------------------|
| 0    | —                           | —                      | \$10,000                        |
| 1    | (0.2)\$10,000               | \$2,000                | 8,000                           |
| 2    | (0.4)\$8,000                | 3,200                  | 4,800                           |
| 3    | (0.4)\$4,800                | 1,920                  | 2,880                           |
| 4    | \$2,880/2.5 years           | 1,152                  | 1,728                           |
| 5    | \$2,880/2.5 years           | 1,152                  | 576                             |
| 6    | (0.5)\$2,880/2.5 years      | 576                    | 0                               |

# MACRS Schedule

| RECOVERY YEAR | PROPERTY CLASS |         |         |         |
|---------------|----------------|---------|---------|---------|
|               | 3-YEAR         | 5-YEAR  | 7-YEAR  | 10-YEAR |
| 1             | 33.33%         | 20.00%  | 14.29%  | 10.00%  |
| 2             | 44.45          | 32.00   | 24.49   | 18.00   |
| 3             | 14.81          | 19.20   | 17.49   | 14.40   |
| 4             | 7.41           | 11.52   | 12.49   | 11.52   |
| 5             |                | 11.52   | 8.93    | 9.22    |
| 6             |                | 5.76    | 8.92    | 7.37    |
| 7             |                |         | 8.93    | 6.55    |
| 8             |                |         | 4.46    | 6.55    |
| 9             |                |         |         | 6.56    |
| 10            |                |         |         | 6.55    |
| 11            |                |         |         | 3.28    |
| Totals        | 100.00%        | 100.00% | 100.00% | 100.00% |

# MACRS Questions

- Bernstein Tractor Company has just invested in new equipment costing \$16,000. The equipment falls in the five-year property class for cost recovery (depreciation) purposes. What depreciation charges can it claim on the asset for each of the next six years?
- The Loann Le Milling Company is going to purchase a new piece of testing equipment for \$28,000 and a new machine for \$53,000. The equipment falls in the three-year property class, and the machine is in the five-year class. What annual depreciation will the company be able to take on the two assets?

# Interest Expense versus Dividends Paid.

- Interest paid on outstanding corporate debt is treated as an expense and is tax deductible.
- However, dividends paid to preferred or common stockholders are not tax deductible.
- Thus, for a profitable, tax-paying company, the use of debt (e.g., bonds) in its financing mix results in a significant tax advantage relative to the use of preferred or common stock.

# Dividend Income

- Dividend Income. A corporation may own stock in another company. If it receives a cash dividend on this stock, normally 70 percent of the dividend is tax exempt.
- The tax laws allow this tax break for corporations (not individuals) to help reduce the effects of multiple taxation of the same earnings.

# Handling Corporate Losses and Gains

- Carryback and Carryforward. If a corporation sustains a net operating loss, this loss may generally be carried back 2 years and forward up to 20 years to offset taxable income in those years.
- Any loss carried back must first be applied to the earliest preceding year.



# Handling Corporate Losses and Gains

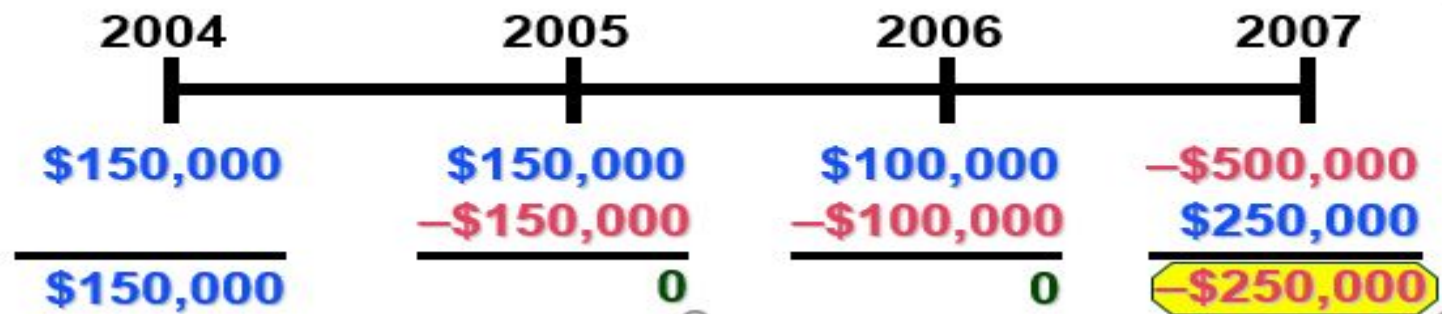
- If a firm sustained an operating loss of \$400,000 in 2008 it would first carry this loss back to 2006.
- If the company had net profits of \$400,000 in that year and paid taxes of \$136,000, it would recompute its taxes for 2006 to show zero profit for tax purposes. Consequently, the company would be eligible for a tax refund of \$136,000.
- If the 2008 operating loss was greater than operating profits in 2006, the residual would be carried back to 2007 and taxes recomputed for that year.

# Handling Corporate Losses and Gains

- Lisa Miller is examining the impact of an operating loss at Basket Wonders (BW) in 2007. The following time line shows operating income and losses. The Tax Rate is 29% What impact does the 2007 loss have on BW?

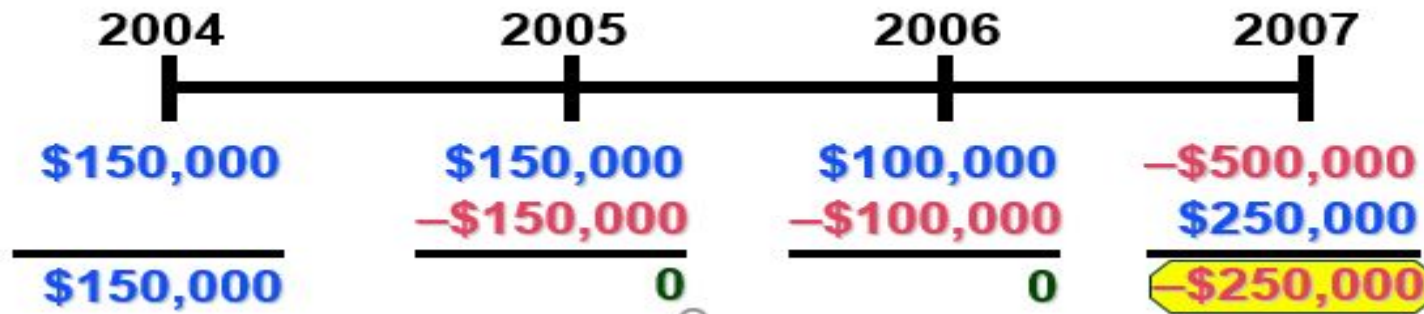


- The loss can offset the gain in each of the years 2005 and 2006. The remaining \$250,000 can be carried forward to 2008 or beyond.



# Handling Corporate Losses and Gains

- The loss can offset the gain in each of the years 2005 and 2006. The remaining **\$250,000** can be carried forward to 2008 or beyond.



| Year                   | 2004     | 2005      | 2006      | 2007      | 2008      |
|------------------------|----------|-----------|-----------|-----------|-----------|
| Amount                 | 150,000  | 150,000   | 100,000   | (500,000) | 300,000   |
| Loss Recovered         |          | (150,000) | (100,000) | 250,000   | (250,000) |
| Remaining Amount       | 150,000  | -         | -         | (250,000) | 50,000    |
| Tax Refund / (Payable) | (43,500) | 43,500    | 29,000    | -         | (14,500)  |

# Capital Gains and Losses

- Capital gain (loss) The amount by which the proceeds from the sale of a capital asset exceeds (is less than) the asset's original cost.
- When a capital asset is sold, a **capital gain or loss** is generally incurred. Often in the history of our tax laws, there has been a differential tax treatment of capital gains income and operating income, with capital gains being treated more favorably.
- Under the Revenue Reconciliation Act of 1993, however, capital gains are taxed at the ordinary income tax rates for corporations, or a maximum of 35 percent.
- Capital losses are deductible only against capital gains.

# Personal Income Taxes

- The US has a progressive tax structure with four tax brackets of 10%, 15%, 25%, 28%, 33%, and 35%.
- The current maximum cash dividend (most) and capital gains tax rates is 15%.
- Personal income taxes are determined by taxable income, filing status, and various credits.
- Result is that low income individuals pay no federal tax and others may fluctuate between the marginal rates.
- Interest, Dividends, and Capital Gains. For the individual, interest received on corporate and Treasury securities is fully taxable at the federal level.

# The Financial Environment

- Businesses interact continually with the **financial markets**.
- *Financial Markets* are composed of all institutions and procedures for bringing buyers and sellers of financial instruments together.
- most firms use financial markets to help finance their investment in assets.
- In the final analysis, the market price of a company's securities is the test of whether the company is a success or a failure.

# The Purpose of Financial Markets

- Financial assets exist in an economy because the savings of various individuals, corporations, and governments during a period of time differ from their investment in real assets.
- A financial asset is created only when the investment of an economic unit in real assets exceeds its savings, and it finances this excess by borrowing or issuing stock.
- The purpose of financial markets in an economy is to allocate savings efficiently to ultimate users.

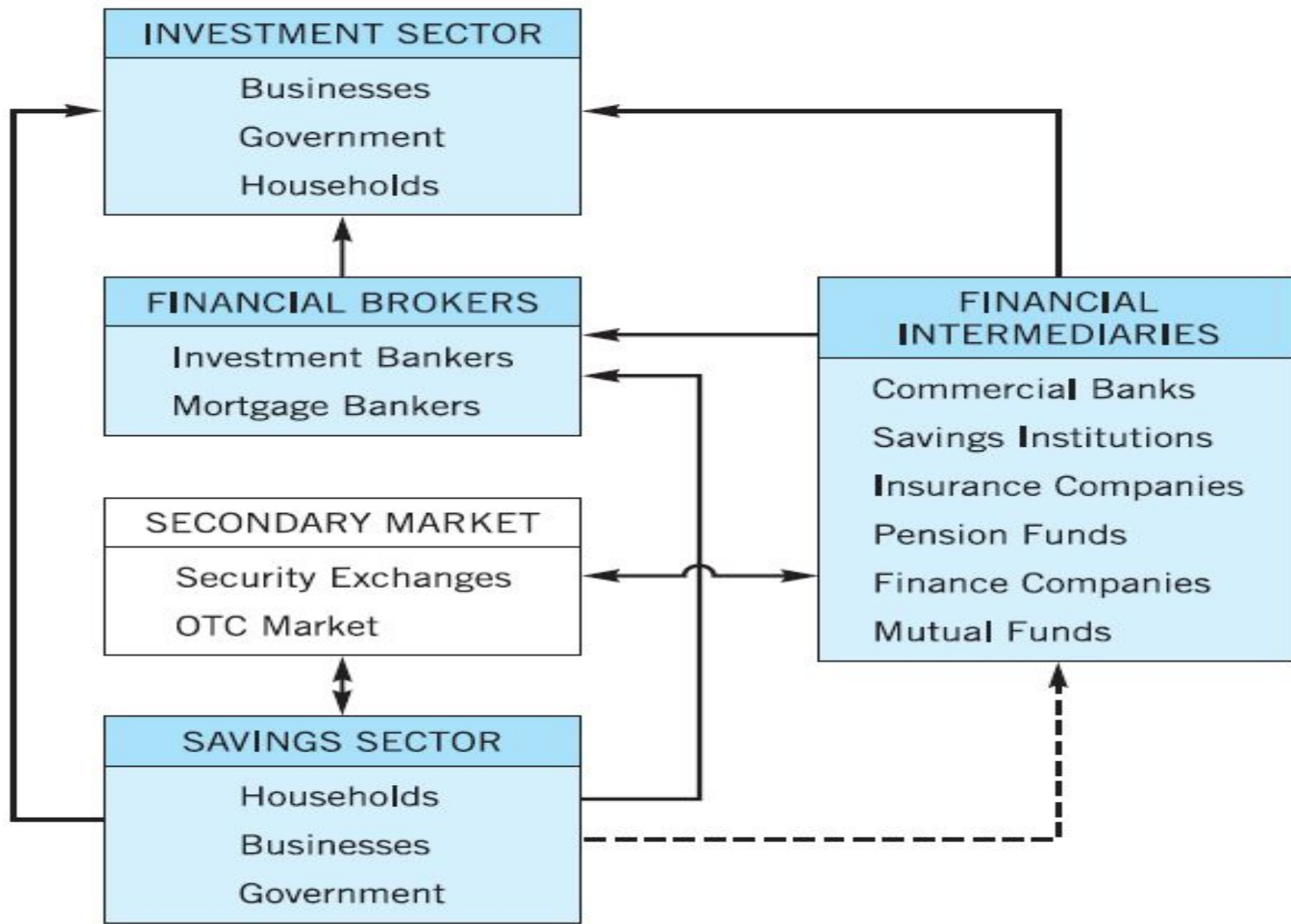
# Financial Markets

- Financial markets are not so much physical places as they are mechanisms for channeling savings to the ultimate investors in real assets.
- There is transfer to funds from savings sector (savings-surplus units) to the investment sector (savings-deficit units).
- Financial markets can be broken into two classes – the money market and the capital market.
- The **money market** is concerned with the buying and selling of short-term (less than one year original maturity) government and corporate debt securities.
- The **capital market**, on the other hand, deals with relatively long-term (greater than one year original maturity) debt and equity instruments (e.g., bonds and stocks).



# Primary and Secondary Markets.

- Within money and capital markets there exist both primary and secondary markets.
- A **primary market** is a “new issues” market. Here, funds raised through the sale of new securities flow from the ultimate savers to the ultimate investors in real assets.
- In a **secondary market**, existing securities are bought and sold. Transactions in these already existing securities do not provide additional funds to finance capital investment.
- With a viable secondary market, a purchaser of financial securities achieves marketability.
- If the buyer needs to sell a security in the future, he or she will be able to do so. Thus, the existence of a strong secondary market enhances the efficiency of the primary market.



# Financial Intermediaries

- The flow of funds from savers to investors in real assets can be direct; if there are financial intermediaries in an economy, the flow can also be indirect.
- **Financial intermediaries** consist of financial institutions, such as commercial banks, savings institutions, insurance companies, pension funds, finance companies, and mutual funds.
- These intermediaries come between ultimate borrowers and lenders by transforming direct claims into indirect claims.
- Financial intermediaries purchase *direct* (or *primary*) *securities* and, in turn, issue their own *indirect* (or *secondary*) *securities* to the public.

# Financial Intermediaries

- *Financial intermediation* is the process of savers depositing funds with financial intermediaries (rather than directly buying stocks and bonds) and letting the intermediaries do the lending to the ultimate investors.

# Example of Financial Intermediaries

- **Deposit Institutions.** *Commercial banks* are the most important source of funds for business firms in the aggregate. Banks acquire demand (checking) and time (savings) deposits from individuals, companies, and governments and, in turn, make loans and investments.
- **Insurance Companies.** There are two types of insurance companies: property and casualty companies and life insurance companies. These are in the business of collecting periodic payments from those they insure in exchange for providing payouts should events, usually adverse, occur.
- With the funds received in premium payments, insurance companies build reserves. These reserves and a portion of the insurance companies' capital are invested in financial assets.

# Example of Financial Intermediaries

- ***Pension funds*** and other ***retirement funds*** are established to provide income to individuals when they retire. During their working lives, employees usually contribute to these funds, as do employers. Funds invest these contributions and either pay out the cumulative amounts periodically to retired workers or arrange annuities. In the accumulation phase, monies paid into a fund are not taxed
- ***Mutual investment funds*** also invest heavily in corporate stocks and bonds. These funds accept monies contributed by individuals and invest them in specific types of financial assets. The mutual fund is connected with a management company, to which the fund pays a fee (frequently 0.5 percent of total assets per annum) for professional investment management.
- ***Finance companies*** make consumer installment loans, personal loans, and secured loans to business enterprises. These companies raise capital through stock issues as well as through borrowings, some of which are long term but most of which come from commercial banks. In turn, the finance company makes loans.

# Financial Brokers

- Certain financial institutions perform a necessary brokerage function. When brokers bring together parties who need funds with those who have savings, they are not performing a direct lending function but rather are acting as matchmakers, or middlemen.
- **Investment bankers** are middlemen involved in the sale of corporate stocks and bonds. When a company decides to raise funds, an investment banker will often buy the issue (at wholesale) and then turn around and sell it to investors (at retail).
- Investment bankers are continually in the business of matching users of funds with suppliers, they can sell issues more efficiently than can the issuing companies

# Financial Brokers

- **Mortgage bankers** are involved in acquiring and placing mortgages. These mortgages come either directly from individuals and businesses or, more typically, through builders and real estate agents.
- In turn, the mortgage banker locates institutional and other investors for the mortgages.



# The Secondary Market

- Various security exchanges and markets facilitate the smooth functioning of the financial system. Purchases and sales of existing financial assets occur in the *secondary market*.
- Transactions in this market do not increase the total amount of financial assets outstanding, but the presence of a viable secondary market increases the liquidity of financial assets and therefore enhances the primary or direct market for securities.
- *Organized exchanges*, such as the New York Stock Exchange, the American Stock Exchange, and the New York Bond Exchange. the forces of supply and demand determine price in the exchange to improve efficiency in matching.

# The Secondary Market

- The *over-the-counter (OTC) market* serves as part of the secondary market for stocks and bonds not listed on an exchange as well as for certain listed securities. It is composed of brokers and dealers who stand ready to buy and sell securities at quoted prices.

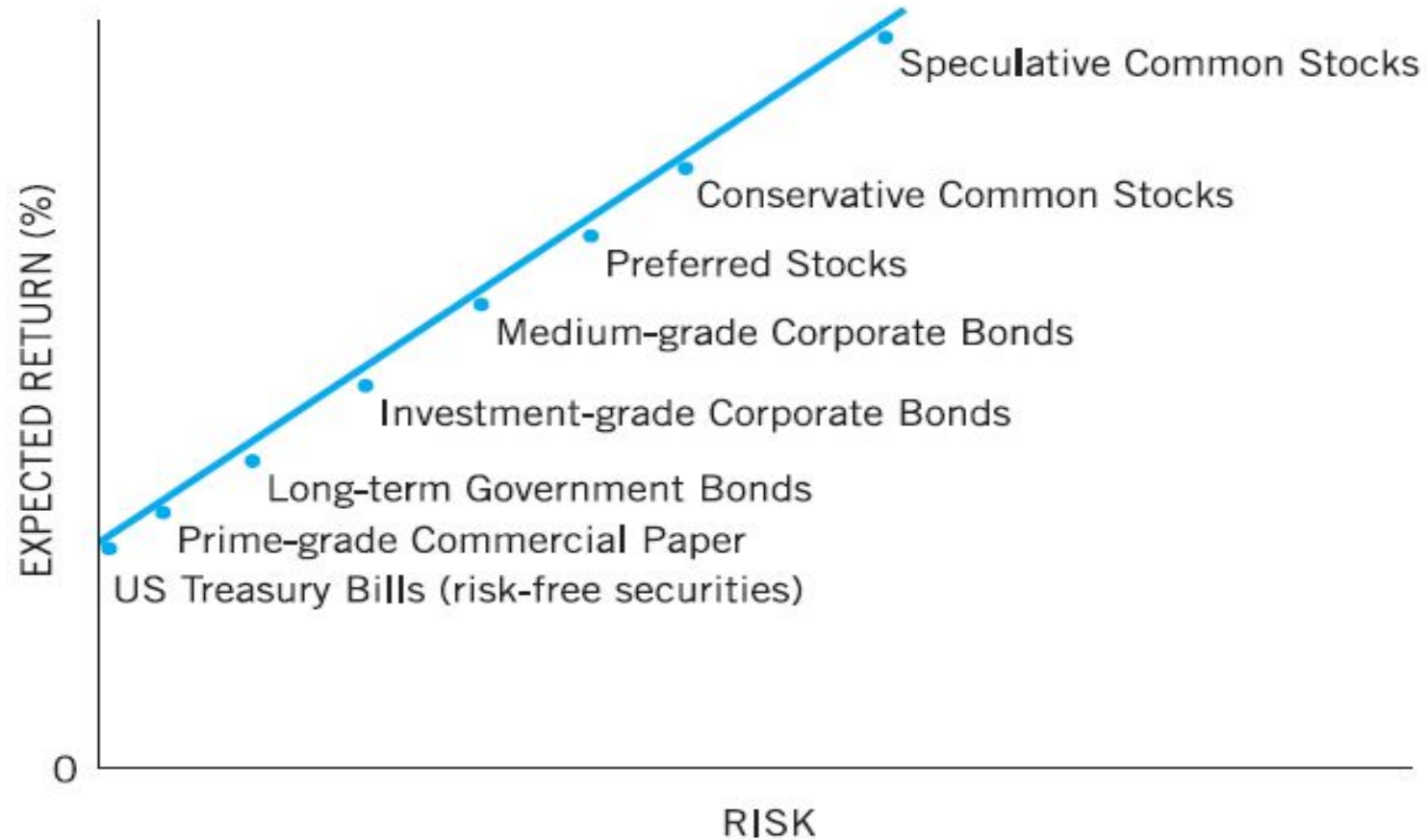
# Allocation of Funds and Interest Rates

- The allocation of funds in an economy occurs primarily on the basis of price, expressed in terms of expected return.
- The allocation process is affected by capital rationing, government restrictions, and institutional constraints, expected return constitutes the primary mechanism by which supply and demand are brought into balance for a particular financial instrument across financial markets.
- If risk is held constant, economic units willing to pay the highest expected return are the ones entitled to the use of funds.
- If people are rational, the economic units bidding the highest prices will have the most promising investment opportunities.
- As a result, savings will tend to be allocated to the most efficient uses.

# Allocation of Funds and Interest Rates

- It is important to recognize that the process by which savings are allocated in an economy occurs not only on the basis of expected return but on the basis of risk as well.
- Different financial instruments have different degrees of risk. In order for them to compete for funds, these instruments must provide different expected returns, or yields.
- The idea of the market-imposed “trade-off ” between risk and return for securities – that is, the higher the risk of a security, the higher the expected return that must be offered to the investor.

# Risk–expected return profile for securities



# What Influences Security Expected Returns?

- **Default Risk.** we mean the danger that the borrower may not meet payments due on principal or interest. Investors demand a risk premium (or extra expected return) to invest in securities that are not default free.
- **Default:** The failure to meet the terms of a contract, such as failure to make interest or principal payments when due on a loan.
- The greater the possibility that the borrower will default, the greater the default risk and the premium demanded by the marketplace.
- For the typical investor, default risk is not judged directly but rather in terms of quality ratings assigned by the principal rating agencies, Moody's Investors Service and Standard & Poor's.
- In their ratings, the agencies attempt to rank issues in order of the perceived probability of default.

# Ratings by Investment Agencies on Default Risk

| MOODY'S INVESTORS SERVICE |                                                        | STANDARD & POOR'S |                           |
|---------------------------|--------------------------------------------------------|-------------------|---------------------------|
| Aaa                       | Best quality                                           | AAA               | Highest grade             |
| Aa                        | High quality                                           | AA                | High grade                |
| A                         | Upper medium grade                                     | A                 | Higher medium grade       |
| Baa                       | Medium grade                                           | BBB               | Medium grade              |
| <hr/>                     |                                                        | <hr/>             |                           |
| Ba                        | Possess speculative elements                           | BB                | Speculative               |
| B                         | Generally lack characteristics of desirable investment | B                 | Very speculative          |
| Caa                       | Poor standing; may be in default                       | CCC–CC            | Outright speculation      |
| Ca                        | Highly speculative; often in default                   | C                 | Bankruptcy petition filed |
| C                         | Lowest grade                                           | D                 | In payment default        |

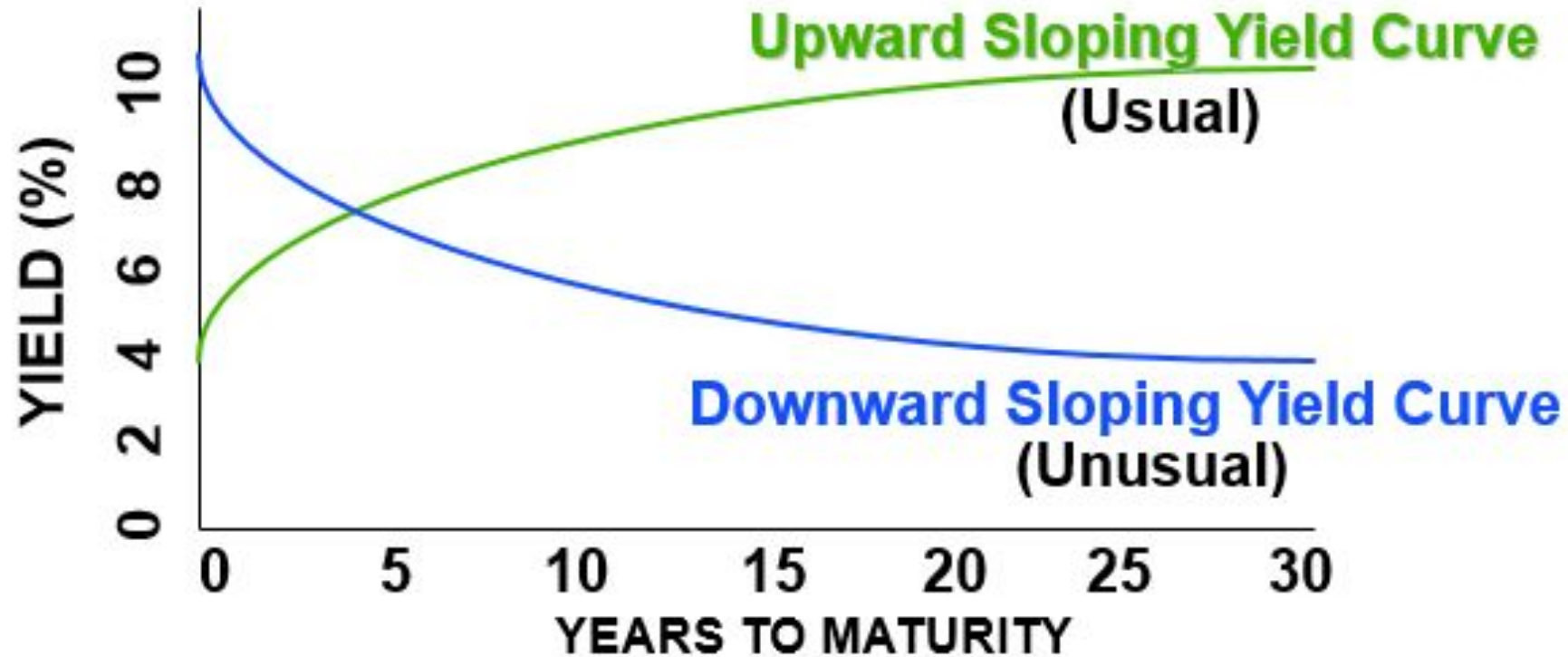
*Note:* The top four categories indicate “investment grade quality” securities; the categories below the dashed line are reserved for securities below investment grade.

# What Influences Expected Security Returns?

- Marketability. The **marketability** (or **liquidity**) of a security relates to the owner's ability to convert it into cash. There are two dimensions to marketability: the price realized and the amount of time required to sell the asset.
- Maturity: Maturity The life of a security; the amount of time before the principal amount of a security becomes due.
- Securities with about the same default risk, having similar marketability, and not faced with different tax implications can still trade at different yields. Why? "Time" is the answer.
- The **maturity** of a security can often have a powerful effect on expected return, or yield.
- The relationship between yield and maturity for securities *differing only in the length of time (or term) to maturity* is called the **term structure of interest rates**.
- The graphical representation of this relationship at a moment in time is called a **yield curve**.



# Term Structure of Interest Rates



# What Influences Expected Security Returns?

- **Taxability.** Another factor affecting the observed differences in market yields is the differential impact of taxes.
- **Option Features.** Another consideration is whether a security contains any option features, such as a conversion privilege or warrants, which upon exercise allow the investor to obtain common stock.
- Other options include the call feature, which enables a company to prepay its debt, and a sinking-fund provision, which allows a company to retire bonds periodically with cash payments or by buying bonds in the secondary market.

# What Influences Expected Security Returns?

- **Inflation.** In addition to the preceding factors, which affect the yield of one security relative to that of another, inflation expectations have a substantial influence on interest rates overall.
- It is generally agreed that the nominal (observed) rate of interest on a security embodies a premium for inflation. The higher the expected inflation, the higher the nominal yield on the security; and the lower the expected inflation, the lower the nominal yield.

# What Influences Expected Security Returns?

- **Behavior of Yields on Corporate Securities.** Differences in default risk, marketability, maturity, taxability, and option features affect the yield of one security relative to another at a point in time.
- In addition, the security yields themselves (and hence the cost of funds to business firms) will vary over time. Fluctuations in supply and demand pressures in financial markets, as well as changing inflation expectations, help explain this variability in yields.